

Dart Learning Path

Phase 1: Foundations (Week 1-2)

Getting Started

- Install Dart SDK and set up your development environment
- Understand what Dart is and its use cases (Flutter, web, server-side)
- Write your first "Hello World" program
- Learn to use DartPad for quick experimentation

Basic Syntax & Data Types

- Variables (`var`, `final`, `const`)
- Built-in types: numbers, strings, booleans, lists, sets, maps
- Type inference and type annotations
- String interpolation and multi-line strings
- Comments and documentation

Control Flow

- if-else statements and conditional expressions
- switch-case statements
- for, while, and do-while loops
- break and continue statements
- assert statements

Phase 2: Core Concepts (Week 3-4)

Functions

- Function declarations and expressions
- Parameters: positional, named, optional, default values
- Arrow syntax for single-expression functions
- Anonymous functions and lambdas
- First-class functions and higher-order functions

Collections